

Replication Project

Who Shot the Bullets? Exposure to Violence and Attitudes Toward
Peace: Evidence from the 2016 Colombian Referendum

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Applied Stats II
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1. Original Study

The study I have chosen to replicate is "Kreiman, G., & Masullo, J. (2020). Who Shot the Bullets? Exposure to Violence and Attitudes Toward Peace: Evidence from the 2016 Colombian Referendum. *Latin American Politics and Society*, 62(4), 24–49. doi:10.1017/lap.2020.14".

This study explores a potential reasoning behind the surprising results of the 2016 Peace Referendum in Colombia. The referendum was brought to a public vote in 2016 by the Colombian president Juan Manuel Santos, following peace talks and a subsequent ceasefire agreement between the Colombian government and the Revolutionary Armed Forces of Colombia (referred to as "FARC" from this point forward). FARC were known as the largest and most powerful guerilla group operating throughout the 50 years of the Colombian civil war. The referendum also included provisions for rural reform, strategies to reduce coca growth and drug trafficking, as well as the allowance of FARC to participate in elections. The public vote resulted in the referendum not being passed by an incredibly small margin. Santos and FARC subsequently continued negotiations and Santos passed the referendum through Colombian congress in late 2016.

Due to the surprising results of the public referendum vote, researchers began to explore why this might have been the case. Preliminary research determined that there were clear geographical patterns within the votes, which was also a characteristic of patterns of violence throughout the war. Subsequent hypotheses have stated that support for the referendum was higher in areas more strongly impacted by war-time violence, as well as that areas under FARC control voted against the referendum due to the fact that it did not appear to punish FARC enough. Ultimately, there have been many conflicting beliefs as to why the referendum did not go through on the public vote. Kreiman and Masullo discovered a gap in the research regarding how variation in people's experiences of violence might have impacted their attitudes towards peace, and ultimately the way they voted in the referendum. They hypothesized that the identity of the perpetrator matters in how people are affected by violence and their subsequent attitude towards peace. Specifically, that individuals in municipalities more impacted by FARC violence were more supportive of the referendum, whereas people more impacted by paramilitary violence were less supportive. The main paramilitary group considered in this investigation was BACRIM (criminal bands in spanish), a paramilitary group formed following the demobilization of the United Self Defence Forces of Colombia (AUC). The AUC was a paramilitary group formed in the 1980s with the intention of combatting the advancements of FARC.

Hypotheses:

- In areas where Group A has committed most of the violence, residents are more likely to support a peace agreement with Group A.
 - Specifically, individuals living in areas more impacted by FARC violence, were more likely to vote in support of the peace agreement with the FARC.
- In areas where B has committed most of the violence, residents are less likely to support

a peace agreement with Group A.

- Specifically, individuals living in areas more impacted by paramilitary/BACRIM violence, were more likely to vote in opposition of the peace agreement with the FARC.

Method:

In order to determine how violence perpetrated by FARC and/or paramilitaries/BACRIM impacted the victims' votes in the 2016 referendum, Kreiman and Masullo gathered previously collected data on overall exposure to violence in municipalities throughout Colombia, number of attacks perpetrated by each group in each municipality, the percentage of votes per municipality supporting the referendum, as well as various control variables. This was a quantitative study utilizing an ordinary least squares regression with departmental fixed effects to make inferences regarding their research question and hypotheses.

2. Data

```
1 # load dataset
2 df_all <- read.dta13("Kreiman_Masullo_Replication_LAPS_Final.dta")
3
4 # remove IVs that are not being used in regression and figures
5 df <- subset(df_all, select = -c(16:26, 28:44)) # columns 16–26 and 28–44 are
6                                              # not used in the regression
```

Outcome Variable:

- % of votes per municipality supporting the 2016 peace agreement referendum

Independent Variables:

Set of variables measuring the average number of violent attacks carried out by FARC and paramilitaries/BACRIM from the years 1988-2010.

- (ln) FARC attacks (ln_atfarc_cede)
- (ln) Paramilitary/BACRIM attacks (ln_atparabacrim_cede)

Control Variables:

Set of variables included in their models that could have an impact on perceived violence and voting in referendum, including political influences, socioeconomic and demographic influences, and variability of natural resources.

- % Votes for Santos in 2014 Election (ref_partic)
- % Participation in Referendum (share_santos_2_100)
- Poverty Measure (incidence_multidim_pov_2005_100)

- Population Measure (lpop)
- Rurality Index (indrural2005)
- Coca Cultivation (cultivated100)
- Oil Present (oil)
- Elevation (ln_altitud)
- Education (coberturabrutaeduc)

Fixed Effects:

A fixed effects model was utilized to control for unobserved characteristics that do not change over time and are specific to each department, as Colombia consists of 32 departments and one capital district (Bogota). For the purpose of departmental fixed effects, Bogota was considered its own department. This model added a dummy variable for each department, which were not represented within the final model, but were included in the regression to absorb the effects of department-level differences between municipalities.

3. Original Model

The authors utilized Stata, which has a wildcard variable denotation to match naming patterns that the authors used to capture the 33 departments in their formula. As R does not have this capability, I created an object that included all of the departments that I pasted onto each formula.

```
1 # create object that allows the departamentos to be easily placed at the end
  of each formula
2 all_depts <- grep("^departamento_", names(df), value = TRUE)
```

There is an incredibly important note that must be made prior to the presentation of the replication code. The authors did not use a consistent reference category for the department dummy variable. Stata typically utilizes the same default reference category as R, which is the lowest-valued category, but upon replicating the regression this did not appear to be the case. For this reason, I created a loop that determined the reference category for each of the 7 models that were run. This loop compared the intercept value obtained using each reference category to the one presented for that model by the authors until it found the reference category that resulted in an intercept within 0.01 of that in the paper.

Model 1: FARC Attacks Only

```
1 # find which reference category matches stata's intercept of 53.67 for model 1
2 for (ref in all_depts) { # iterate over each department dummy
3   dept_vars_try <- paste(setdiff(all_depts, ref), collapse = " + ") # build
  formula without the current department dummy
```

```

4 f_try <- as.formula(paste("ref_si ~ ln_atfarc_cede +", dept_vars_try)) #
  build full formula
5 model_try <- lm(f_try, data = df) # try model without dummy
6 intercept <- coef(model_try)["(Intercept)"] # isolate intercept
7 if (abs(intercept - 53.67) < 0.01) {
8   cat("Model 1 reference category:", ref, "\n") # if the intercept is within
  0.01, this is the reference category
9 }
10 }
11
12 # remove reference department from all departments
13 dept_vars_1 <- paste(setdiff(all_depts, "departamento_27"), collapse = " + ")
14
15 # run model 1
16 f1 <- as.formula(paste("ref_si ~ ln_atfarc_cede +", dept_vars_1)) # formula
17 model_1 <- lm(f1, data = df) # run regression
18 summary(model_1) # check regression coefficients are accurate

```

Model 2: Para/BACRIM Attacks Only

```

1 # find which reference category matches stata's intercept for model 2
2 for (ref in all_depts) {
3   dept_vars_try <- paste(setdiff(all_depts, ref), collapse = " + ")
4   f_try <- as.formula(paste("ref_si ~ ln_atparabacrim_cede +", dept_vars_try))
5   model_try <- lm(f_try, data = df)
6   intercept <- coef(model_try)["(Intercept)"]
7   if (abs(intercept - 60.89) < 0.01) {
8     cat("Model 2 reference category:", ref, "\n")
9   }
10 }
11
12 # remove reference department from all departments
13 dept_vars_2 <- paste(setdiff(all_depts, "departamento_5"), collapse = " + ")
14
15 # run model 2
16 f2 <- as.formula(paste("ref_si ~ ln_atparabacrim_cede +", dept_vars_2))
17 model_2 <- lm(f2, data = df)
18 summary(model_2)

```

Model 3: FARC and Para/BACRIM Attacks Only

```

1 # find which reference category matches stata's for model 3
2 for (ref in all_depts) {
3   dept_vars_try <- paste(setdiff(all_depts, ref), collapse = " + ")
4   f_try <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +",
  , dept_vars_try))
5   model_try <- lm(f_try, data = df)
6   intercept <- coef(model_try)["(Intercept)"]
7   if (abs(intercept - 69.81) < 0.01) {
8     cat("Model 3 reference category:", ref, "\n")
9   }
10 }
11

```

```

12 # remove reference department from all departments
13 dept_vars_3 <- paste(setdiff(all_depts, "departamento_32"), collapse = " + ")
14
15 # run model 3
16 f3 <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +",
17   dept_vars_3))
18 model_3 <- lm(f3, data = df)
19 summary(model_3)

```

Model 4: FARC Attacks, Para/BACRIM Attacks, % Participation in Referendum, and % Support for Santos in 2014

```

1 # find which reference category matches stata's for model 4
2 for (ref in all_depts) {
3   dept_vars_try <- paste(setdiff(all_depts, ref), collapse = " + ")
4   f_try <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
5     ref_partic + share_santos_2_100 +", dept_vars_try))
6   model_try <- lm(f_try, data = df)
7   intercept <- coef(model_try)["(Intercept)"]
8   if (abs(intercept - 35.80) < 0.01) {
9     cat("Model 4 reference category:", ref, "\n")
10  }
11 }
12
13 # remove reference department from all departments
14 dept_vars_4 <- paste(setdiff(all_depts, "departamento_5"), collapse = " + ")
15
16 # run model 4
17 f4 <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
18   ref_partic + share_santos_2_100 +", dept_vars_4))
19 model_4 <- lm(f4, data = df)
20 summary(model_4)

```

Model 5: FARC Attacks, Para/BACRIM Attacks, % Participation in Referendum, % Support for Santos in 2014, Poverty, Population, and Rural Index

```

1 # find which reference category matches stata's for model 5
2 for (ref in all_depts) {
3   dept_vars_try <- paste(setdiff(all_depts, ref), collapse = " + ")
4   f_try <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
5     ref_partic + share_santos_2_100 +
6     incidence_multidim_pov_2005_100 + lpop + indrural2005 +
7     ", dept_vars_try))
8   model_try <- lm(f_try, data = df)
9   intercept <- coef(model_try)["(Intercept)"]
10  if (abs(intercept - 30.22) < 0.01) {
11    cat("Model 5 reference category:", ref, "\n")
12  }
13 }
14
15 # remove reference department
16 dept_vars_5 <- paste(setdiff(all_depts, "departamento_5"), collapse = " + ")

```

```

17 # run model 5
18 f5 <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
19                       ref_partic + share_santos_2_100 +
20                       incidence_multidim_pov_2005_100 + lpop + indrural2005 +
21                       ",
22                       dept_vars_5))
23 model_5 <- lm(f5, data = df)
24 summary(model_5)

```

Model 6: FARC Attacks, Para/BACRIM Attacks, % Participation in Referendum, % Support for Santos in 2014, Poverty, Population, Rural Index, Coca, and Oil

```

1 # find which reference category matches stata's for model 6
2 for (ref in all_depts) {
3   dept_vars_try <- paste(setdiff(all_depts, ref), collapse = " + ")
4   f_try <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
5                             ref_partic + share_santos_2_100 +
6                             incidence_multidim_pov_2005_100 + lpop + indrural2005 +
7                             cultivated100 + oil +", dept_vars_try))
8   model_try <- lm(f_try, data = df)
9   intercept <- coef(model_try)[ "(Intercept)" ]
10  if (abs(intercept - 20.01) < 0.01) {
11    cat("Model 6 reference category:", ref, "\n")
12  }
13 }
14
15 # remove reference department
16 dept_vars_6 <- paste(setdiff(all_depts, "departamento_3"), collapse = " + ")
17
18 # run model 6
19 f6 <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
20                       ref_partic + share_santos_2_100 +
21                       incidence_multidim_pov_2005_100 + lpop + indrural2005 +
22                       cultivated100 + oil +", dept_vars_6))
23 model_6 <- lm(f6, data = df)
24 summary(model_6)

```

Model 7: FARC Attacks, Para/BACRIM Attacks, % Participation in Referendum, % Support for Santos in 2014, Poverty, Population, Rural Index, Coca, Oil, Elevation, and Education Coverage

```

1 # find which reference category matches stata's for model 7
2 for (ref in all_depts) {
3   dept_vars_try <- paste(setdiff(all_depts, ref), collapse = " + ")
4   f_try <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
5                             ref_partic + share_santos_2_100 +
6                             incidence_multidim_pov_2005_100 + lpop + indrural2005 +
7                             cultivated100 + oil + ln_altitud +
8                             coberturabrutaeduc +", dept_vars_try))
9   model_try <- lm(f_try, data = df)
10  intercept <- coef(model_try)[ "(Intercept)" ]

```

Table 1: OLS Identity Perpetrator (Models 1–4)

	(1)	(2)	(3)	(4)
(ln) FARC attacks	0.785*** (0.277)		1.835*** (0.321)	1.302*** (0.243)
(ln) Paramilitary/BACRIM attacks		-1.166*** (0.322)	-2.151*** (0.385)	-2.400*** (0.289)
% Participation in Referendum				-0.090* (0.047)
% Support Santos 2014				0.586*** (0.022)
Department Fixed Effects	✓	✓	✓	✓
Constant	53.671*** (12.487)	60.894*** (12.472)	69.814*** (12.279)	35.796*** (9.512)
Num.Obs.	993	1028	945	944
R2	0.478	0.463	0.481	0.711
R2 Adj.	0.460	0.445	0.462	0.700

* p < 0.1, ** p < 0.05, *** p < 0.01

```

11 if (abs(intercept - 18.78) < 0.01) {
12   cat("Model 7 reference category:", ref, "\n")
13 }
14 }
15
16 # remove reference department
17 dept_vars_7 <- paste(setdiff(all_depts, "departamento_24"), collapse = " + ")
18
19 # run model 7
20 f7 <- as.formula(paste("ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
21   ref_partic + share_santos_2_100 +
22   incidence_multidim_pov_2005_100 + lpop + indrural2005 +
23   cultivated100 + oil + ln_altitud +
24   cobeturabrutaeduc +", dept_vars_7))
25 model_7 <- lm(f7, data = df)
26 summary(model_7)

```

The authors stated that these results supported their hypothesis, illustrating that the identity of the perpetrator was an influence in the 2016 referendum as the impact of (ln) FARC attacks and (ln) Para/BACRIM Attacks on the % yes vote in the referendum appeared to be non-zero and statistically significant. Focusing on the FARC and Paramilitary/BACRIM attacks variables in model 7, these can be interpreted as:

(ln) FARC Attacks: Holding all other variables constant, on average, for every one

Table 2: OLS Identity Perpetrator (Models 5–7)

	(5)	(6)	(7)
(ln) FARC attacks	1.100*** (0.249)	1.050*** (0.254)	1.058*** (0.264)
(ln) Paramilitary/BACRIM attacks	−1.608*** (0.418)	−1.470*** (0.431)	−1.322*** (0.445)
% Participation in Referendum	−0.031 (0.051)	−0.037 (0.053)	−0.069 (0.055)
% Support Santos 2014	0.596*** (0.022)	0.589*** (0.023)	0.585*** (0.025)
Poverty	0.000 (0.035)	0.003 (0.036)	−0.008 (0.040)
(ln) Population	−0.018 (0.527)	−0.126 (0.538)	−0.196 (0.556)
Rural Index	0.092*** (0.021)	0.083*** (0.022)	0.085*** (0.023)
Coca		2.022** (1.028)	2.128** (1.039)
Oil		−1.627 (1.036)	−1.781* (1.047)
(ln) Elevation			0.113 (0.358)
Education Coverage			0.020 (0.019)
Department Fixed Effects	✓	✓	✓
Constant	30.221** (12.045)	20.008*** (7.414)	18.778* (9.602)
Num.Obs.	927	887	857
R2	0.720	0.710	0.699
R2 Adj.	0.708	0.698	0.685

* p < 0.1, ** p < 0.05, *** p < 0.01

percent increase in the mean number of FARC attacks in a municipality, the percentage of votes per municipality supporting the 2016 peace agreement referendum increases by .01058 percentage points.

(ln) Para/BACRIM Attacks: Holding all other variables constant, on average, for every one percent increase in the mean number of Para/BACRIM attacks in a municipality, the percentage of votes per municipality supporting the 2016 peace agreement referendum decreases by .0132 percentage points.

The authors also noted that municipalities with higher support for Santos saw a statistically significant increase in the percentage of votes in favor of the referendum, as well as more rural municipalities and those with higher coca cultivation.

The authors also generated Figure 1, which was created by including variables that captured overall exposure to violence within a municipality through (ln) `exposure total` and (ln) `exposure total per 1,000 people`. These variables represent the log of the number of attacks within a municipality not aggregated by the group perpetrating the violence. To obtain this figure, each overall exposure variable was run separately in an OLS regression that included just the control variables listed above, and excluded the (ln) `FARC attacks` and (ln) `Para/BACRIM attacks` variables. This code was not supplied within the replication code, but I was able to accurately replicate the figure myself using the code below.

```

1 # Model with exposure total mean
2 f8 <- as.formula(paste(c("ref_si ~ ln_exposure_total_mean +
3                           ref_partic + share_santos_2_100 +
4                           incidence_multidim_pov_2005_100 + lpop + indrural2005 +
5                           cultivated100 + oil + ln_altitud +
6                           coberturabrutaeduc +", all_depts), collapse = " + "))
7 model_8 <- lm(f8, data = df_all)
8 summary(model_8)
9
10 # Model with exposure total 1000 pop mean
11 f9 <- as.formula(paste(c("ref_si ~ ln_exposure_total_1000pop_mean +
12                           ref_partic + share_santos_2_100 +
13                           incidence_multidim_pov_2005_100 + lpop + indrural2005 +
14                           cultivated100 + oil + ln_altitud +
15                           coberturabrutaeduc +", all_depts), collapse = " + "))
16 model_9 <- lm(f9, data = df_all)
17 summary(model_9)
18
19 # Make data frame that will be plotted
20 coef_data <- data.frame(
21   variable = c("(ln) Exposure Total",
22                 "(ln) Exposure Total 1000 Pop.",
23                 "(ln) FARC Attacks",
24                 "(ln) Para/BACRIM Attacks"),
25   estimate = c(model_8$coefficients["ln_exposure_total_mean"],
26                 model_9$coefficients["ln_exposure_total_1000pop_mean"],
27                 model_7$coefficients["ln_atfarc_cede"],
28                 model_7$coefficients["ln_atparabacrim_cede"]),
29   se = c(0.34520, 0.34420, 0.263773, 0.445106) # taken from model summaries

```

```

30 ) |>
31   mutate(
32     ci_low = estimate - 1.96 * se,
33     ci_high = estimate + 1.96 * se,
34     significant = ifelse(ci_low > 0 | ci_high < 0, "yes", "no"),
35     variable = factor(variable, levels = c("(ln) Para/BACRIM Attacks",
36                                           "(ln) FARC Attacks",
37                                           "(ln) Exposure Total 1000 Pop.",
38                                           "(ln) Exposure Total"))
39   )
40
41 # create plot
42 pdf("figure_1.pdf")
43 figure_1 <- ggplot(coef_data, aes(x = estimate, y = variable)) +
44   geom_vline(xintercept = 0, linetype = "solid", color = "black") +
45   geom_segment(aes(x = ci_low, xend = ci_high,
46                   y = variable, yend = variable),
47               color = "purple", linewidth = 0.8) +
48   geom_point(size = 3, color = "purple") +
49   labs(
50     x = "DV: Support for the \"Yes\" in the 2016 Referendum",
51     y = NULL
52   ) +
53   coord_cartesian(xlim = c(min(coef_data$ci_low) - 0.2,
54                             max(coef_data$ci_high) + 0.2)) +
55   scale_y_discrete(expand = expansion(mult = 0.15)) +
56   theme_minimal() +
57   theme(
58     legend.position = "none", # remove the legend
59     panel.grid.major.y = element_blank(),
60     panel.grid.minor = element_blank(),
61     axis.text.y = element_text(size = 11),

```

From figure 1, the authors stated that "overall levels of violent attacks by all armed groups in each Colombian municipality do not have an impact on citizens' support for the peace agreement, clear (and statistically significant) effects emerge when we disaggregate by the identity of the perpetrator." These results substantiated their claim that it was important to disaggregate exposure to violence when looking at reasoning behind the 2016 referendum results, as overall exposure to violence was not a significant predictor in support for the referendum.

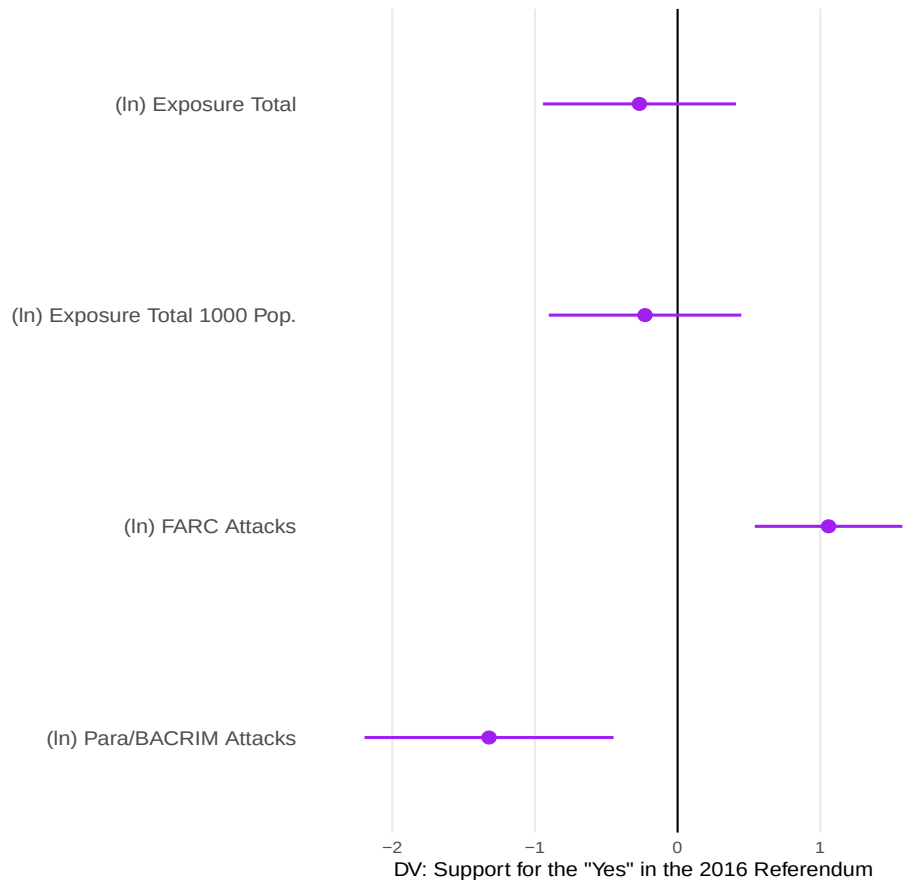


Figure 1: Exposure to Violence and Vote Choices in the Referendum

4. My Contribution

My contribution was spurred by the discovery of non-identical intercept values following replication of the regression code. This led to the realization that the authors did not use a consistent reference category for their department fixed effects dummy variable. After discovering this, I realized that some of the department groups contain very few municipalities within them and therefore have an incredibly small group size. These small, unbalanced group sizes do not allow for estimation of within-group variations and resultingly do not produce precise or accurate estimates. Controlling for fixed effects is still important, as there is variation between departments that may impact the results and need to be controlled for.

In order to do this accurately, I implemented a multilevel model (MLM) to estimate the regression coefficients. MLM is a method that is an extension of linear regression, but allows for more accurate inferences for grouped data by allowing the intercept, or both the intercept and slope, to vary. Both group and individual level variation is considered within the estimation of the group-level coefficients, allowing for accurate inferences to be made for groups that have small sample sizes, which is highly appropriate in this setting. An important concept in MLM that helps explain how it functions is pooling, which refers to the way in which information is shared across groups in a model. A "no pooling" model is one in which no information is shared between groups and the coefficients are estimated purely based on variation within their group. A full pooling model means that groups are not considered, ignoring differences that might arise from between group variation. An MLM is in between these two pooling extremes, allowing group-level estimates to be a weighted combination of within and between group data. A small group size will then pull more data from the overall larger sample average, rather than from the information within its group, which is necessary to obtain accurate estimates from this data (Gelman & Hill, 2017).

For this particular instance, there will be two levels, the department level and the municipality level. A varying intercept model is utilized, allowing the intercept to vary for each group, but keeping the slopes the same.

The first step that I had to do prior to running the models was to create a new column in the dataframe that held all of the department's identities.

```
1 # need to create a new column with the departments together and factor it
2 df$departamento <- factor(
3   names(df[, all_depts])[max.col(df[, all_depts])])
```

Model 1: FARC Attacks Only

```
1 # model 1: FARC attacks only
2 multi_model_1 <- lmer(ref_si ~ ln_atfarc_cede + (1 | departamento), data = df)
3 summary(multi_model_1)
```

Model 2: Para/BACRIM Attacks Only

```
1 # model 2: Para/BACRIM attacks only
2 multi_model_2 <- lmer(ref_si ~ ln_atparabacrim_cede + (1 | departamento),
3   data = df)
4 summary(multi_model_2)
```

Model 3: FARC and Para/BACRIM Attacks Only

```

1 # model 3: farc and para/BACRIM attacks only
2 multi_model_3 <- lmer(ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
3                       (1 | departamento), data = df)
4 summary(multi_model_3)

```

Model 4: FARC Attacks, Para/BACRIM Attacks, % Participation in Referendum, and % Support for Santos in 2014

```

1 # model 4: farc attacks, para/BACRIM attacks, % participation in ref,
2 # % support santos
3 multi_model_4 <- lmer(ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
4                       ref_partic + share_santos_2_100 + (1 | departamento),
5                       data = df)
6 summary(multi_model_4)

```

Model 5: FARC Attacks, Para/BACRIM Attacks, % Participation in Referendum, % Support for Santos in 2014, Poverty, Population, and Rural Index

```

1 # model 5: farc attacks, para/BACRIM attacks, % participation in ref,
2 # % support santos, poverty, population, rural index
3 multi_model_5 <- lmer(ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
4                       ref_partic + share_santos_2_100 +
5                       incidence_multidim_pov_2005_100 +
6                       lpop + indrural2005 + (1 | departamento),
7                       data = df)
8 summary(multi_model_5)

```

Model 6: FARC Attacks, Para/BACRIM Attacks, % Participation in Referendum, % Support for Santos in 2014, Poverty, Population, Rural Index, Coca, and Oil

```

1 # model 6: farc attacks, para/BACRIM attacks, % participation in ref,
2 # % support santos, poverty, population, rural index, coca, oil
3 multi_model_6 <- lmer(ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
4                       ref_partic + share_santos_2_100 +
5                       incidence_multidim_pov_2005_100 +
6                       lpop + indrural2005 + cultivated100 + oil +
7                       (1 | departamento),
8                       data = df)
9 summary(multi_model_6)

```

Model 7: FARC Attacks, Para/BACRIM Attacks, % Participation in Referendum, % Support for Santos in 2014, Poverty, Population, Rural Index, Coca, Oil, Elevation, and Education Coverage

```

1 # model 7: farc attacks, para/BACRIM attacks, % participation in ref,
2 # % support santos, poverty, population, rural index, coca, oil, elevation,
3 # education coverage
4 multi_model_7 <- lmer(ref_si ~ ln_atfarc_cede + ln_atparabacrim_cede +
5                       ref_partic + share_santos_2_100 +
6                       incidence_multidim_pov_2005_100 +
7                       lpop + indrural2005 + cultivated100 + oil + ln_altitud
8                       + coberturabrutaeduc + (1 | departamento),
9                       data = df)

```

```
10 summary(multi_model_7)
```

The resulting coefficients from the multi-level model are similar to those in the original OLS model, indicating accurate representation of the effects of the covariates through the MLM. The FARC attacks and Para/BACRIM attacks coefficients can be interpreted as follows:

(ln) FARC Attacks: Holding all other variables constant, on average, for every one percent increase in the mean number of FARC attacks in a municipality, the percentage of votes per municipality supporting the 2016 peace agreement referendum increases by .0103%.

(ln) Para/BACRIM Attacks: Holding all other variables constant, on average, for every one percent increase in the mean number of Para/BACRIM attacks in a municipality, the percentage of votes per municipality supporting the 2016 peace agreement referendum decreases by .0134%.

The intercept for this model can now be more accurately interpreted. The first part of its interpretation is held in the (Intercept) estimate, whereas the second part can be found through running the `coef()` function in R on the model. The output of this function tells you the intercept for each department.

(Intercept): The average expected value of the percentage of votes per municipality supporting the 2016 peace agreement referendum, across all departments when all covariates are 0 is 20.16%.

Intercepts by Department: In order to display the shifted intercepts, `coef()` can be called in R. I chose to place it in a dataframe to make it easier to view.

```
1 # determine and show shifted intercepts
2 data.frame(coef(multi_model_7)$departamento)[1]
```

Intercept.	
departamento_10	24.27479
departamento_11	21.28139
departamento_12	17.28407
departamento_13	32.82396
departamento_14	22.50164
departamento_15	18.15005
departamento_18	20.42344
departamento_19	18.11978
departamento_2	18.01475
departamento_21	19.44325
departamento_22	22.14029
departamento_23	13.68684
departamento_24	18.61172
departamento_25	14.17935
departamento_26	22.26773
departamento_27	20.77130
departamento_28	18.13276

Table 3: Multi-level Model Output

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
(Intercept)	55.35*** (2.35)	54.52*** (2.16)	54.02*** (2.18)	27.01*** (2.16)	15.63* (6.66)	18.26** (6.88)	20.16* (8.46)
ln_atfare_cede	0.74** (0.27)		1.81*** (0.32)	1.31*** (0.23)	1.05*** (0.24)	0.98*** (0.25)	1.03*** (0.26)
ln_atparabacrim_cede		-1.20*** (0.32)	-2.16*** (0.38)	-2.45*** (0.28)	-1.67*** (0.41)	-1.45*** (0.42)	-1.34** (0.44)
ref_partic				-0.09* (0.04)	-0.02 (0.05)	-0.03 (0.05)	-0.05 (0.05)
share_santos_2_100				0.59*** (0.02)	0.60*** (0.02)	0.59*** (0.02)	0.58*** (0.02)
incidence_multidim_pov_2005_100					0.03 (0.03)	0.03 (0.03)	0.01 (0.04)
lpop					0.20 (0.51)	-0.03 (0.52)	-0.13 (0.54)
indrural2005					0.09*** (0.02)	0.08*** (0.02)	0.08*** (0.02)
cultivated100						1.73 (0.99)	1.85 (1.00)
oil						-1.76 (1.02)	-1.91 (1.03)
ln_altitud							-0.10 (0.34)
coberturaabrutaeduc						0.02 (0.02)	0.02 (0.02)
AIC	7929.72	8189.33	7512.71	6938.02	6768.53	6485.89	6277.62
BIC	7949.32	8209.07	7536.96	6971.97	6816.85	6543.34	6344.16
Log Likelihood	-3960.86	-4090.66	-3751.35	-3462.01	-3374.27	-3230.94	-3124.81
Num. obs.	993	1028	945	944	927	887	857
Num. groups: departamento	33	33	33	33	33	27	26
Var: departamento (Intercept)	165.60	135.69	138.72	25.63	19.71	19.02	19.47
Var: Residual	155.05	153.59	149.73	83.48	78.73	79.81	79.97

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

departamento_29	25.41456
departamento_3	19.46995
departamento_30	17.73418
departamento_31	15.13746
departamento_4	19.48540
departamento_6	21.77997
departamento_7	24.99865
departamento_8	19.32642
departamento_9	18.76078

The intercept for each department can now be determined. There are a few departments that are missing from this output, such as departments 1, 5, 16, 17, 20, 32, and 33, due to the fact that the random effects intercept was estimated to be the same as the fixed effects intercept. This is most likely due to their small group size and subsequent small variance within the group, causing the intercept to be more accurately estimated by the overall intercept (Clark, n.d.).

As an example I will interpret the intercept for **departamento_10**: The average expected value of the percentage of votes per municipality within department 11 supporting the 2016 peace agreement referendum, when all covariates are 0 is 24.27%.

5. Another Contribution/Correction

Upon initially attempting to interpret the coefficients from the classic OLS model, I came across what I believe is a fault in the author's interpretations. Verbatim, the authors interpret the (ln) FARC attacks and (ln) Para/BACRIM attacks coefficients as: "If the mean of FARC attacks in a given municipality increases by 1%, the yes vote in the referendum increases by approximately 1.06 points." and "If the mean of paramilitary/BACRIM attacks in a municipality increase by 1%, the yes vote in the referendum decreases by approximately 1.32 points."

This interpretation is not entirely accurate, as the log-transformation of the raw FARC and Para/BACRIM attacks counts, and subsequently the interpretation of a 1% increase in the mean of these attacks would not allow the authors to then directly interpret the coefficients without transforming them as well. Taking the log of the attack counts requires the *Beta* coefficient value to then be divided by 100, to account for the transformation of x . This would then result in the interpretations being what I reported when presenting the OLS output results. The difference in value when considering and applying this transformation is incredibly large, greatly changing the magnitude of the results.

Following this realization, I chose to determine predicted probabilities of % yes votes in the referendum for varying numbers of FARC attacks and para/BACRIM attacks, and subsequently plot these values against the exponentiated (ln) FARC Attacks and (ln) Para/BACRIM Attacks variable values to get the true raw counts.

To do this, I created a new dataframe by repeating the first row of the original dataframe 100 times to create new data to make my prediction from. I added a column for the de-

partments and held this at 0. For the covariates, I held them all at their mean apart from `ln_atfarc_cede` and `ln_atparabacrim_cede`, for which I used a varying sample of 100 numbers between the min and max values. Following this, I generated my predictions and calculated confidence intervals before plotting the results against the true FARC and para/BACRIM attack counts.

FARC Attacks:

```

1 # repeat the first 100 rows of the original dataframe
2 prediction_data_farc <- df[rep(1, 100), ]
3
4 # fill all department columns with 0
5 dept_cols <- grep("^departamento-", names(prediction_data_farc), value = TRUE)
6 prediction_data_farc[, dept_cols] <- 0
7
8 # fill every other variable, make all values equal to the mean
9 prediction_data_farc$ln_atparabacrim_cede = mean(df$ln_atparabacrim_cede, na.rm = TRUE)
10 prediction_data_farc$ref_partic = mean(df$ref_partic, na.rm = TRUE)
11 prediction_data_farc$share_santos_2_100 = mean(df$share_santos_2_100, na.rm = TRUE)
12 prediction_data_farc$incidence_multidim_pov_2005_100 = mean(df$incidence_multidim_pov_2005_100, na.rm = TRUE)
13 prediction_data_farc$lpop = mean(df$lpop, na.rm = TRUE)
14 prediction_data_farc$indrural2005 = mean(df$indrural2005, na.rm = TRUE)
15 prediction_data_farc$cultivated100 = mean(df$cultivated100, na.rm = TRUE)
16 prediction_data_farc$oil = mean(df$oil, na.rm = TRUE)
17 prediction_data_farc$ln_altitud = mean(df$ln_altitud, na.rm = TRUE)
18 prediction_data_farc$coberturabrutaeduc = mean(df$coberturabrutaeduc, na.rm = TRUE)
19
20 # have FARC attacks variable vary amongst ranges between the min and max values
21 prediction_data_farc$ln_atfarc_cede <- as.numeric(seq(min(df$ln_atfarc_cede, na.rm = TRUE),
22                                                         max(df$ln_atfarc_cede, na.rm = TRUE),
23                                                         length.out = 100))
24
25 # predict new data using model_7 on the prediction data created
26 preds_farc <- predict(model_7, newdata = prediction_data_farc, type = "response",
27                        interval = "confidence")
28
29 # unlist predicted column
30 prediction_data_farc$predicted <- preds_farc[, "fit"]
31
32 # confidence intervals around predicted probabilities
33 prediction_data_farc$lower_95 <- preds_farc[, "lwr"]
34 prediction_data_farc$upper_95 <- preds_farc[, "upr"]
35
36 # plot predicted % Yes votes in the referendum against farc attacks

```

```

37 pdf("pred_plot.pdf")
38 pred_plot <- ggplot(prediction_data_farc, aes(x = exp(ln_atfarc_cede), y =
    predicted)) +
39   geom_ribbon(aes(ymin=lower_95, ymax=upper_95), alpha = 0.2) +
40   geom_point() +
41   geom_rug(data = df, aes(x=exp_atfarc_cede),
42     inherit.aes = FALSE, alpha = 0.4, sides = "b") +
43   labs(y = "Predicted % Yes Votes in Referendum",
44     x = "FARC Attacks",
45     title = "FARC Attacks vs Predicted Votes in Referendum") +
46   theme_minimal()
47 pred_plot

```

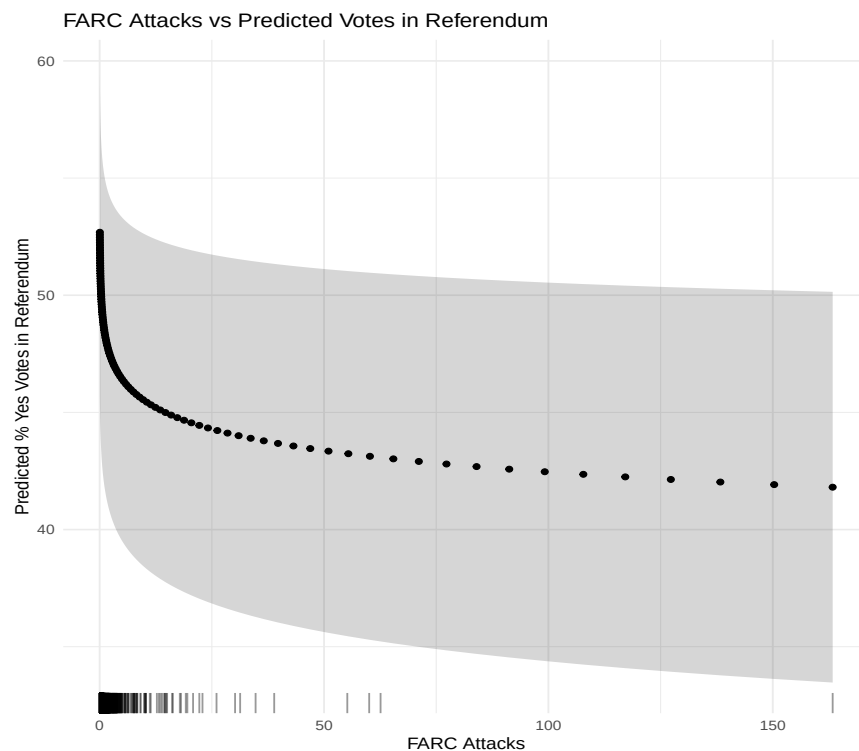


Figure 2: FARC Attack Counts vs Predicted Probabilities of % Yes Votes in Referendum

Para/BACRIM Attacks:

```

1 # repeat the first 100 rows of the original dataframe
2 prediction_data_para <- df[rep(1, 100), ]
3
4 # fill all department columns with 0
5 dept_cols <- grep("^departamento-", names(prediction_data_para), value = TRUE)
6 prediction_data_para[, dept_cols] <- 0
7
8 # fill every other variable, make all values equal to the mean
9 prediction_data_para$ln_atfarc_cede = mean(df$ln_atfarc_cede, na.rm = TRUE)

```

```

10 prediction_data_para$ref_partic = mean(df$ref_partic, na.rm = TRUE)
11 prediction_data_para$share_santos_2_100 = mean(df$share_santos_2_100, na.rm =
    TRUE)
12 prediction_data_para$incidence_multidim_pov_2005_100 = mean(df$incidence -
    multidim_pov_2005_100, na.rm = TRUE)
13 prediction_data_para$lpop = mean(df$lpop, na.rm = TRUE)
14 prediction_data_para$indrural2005 = mean(df$indrural2005, na.rm = TRUE)
15 prediction_data_para$cultivated100 = mean(df$cultivated100, na.rm = TRUE)
16 prediction_data_para$oil = mean(df$oil, na.rm = TRUE)
17 prediction_data_para$ln_altitud = mean(df$ln_altitud, na.rm = TRUE)
18 prediction_data_para$coberturabrutaeduc = mean(df$coberturabrutaeduc, na.rm =
    TRUE)
19
20 # have FARC attacks variable vary amongst ranges between the min and max
    values
21 prediction_data_para$ln_atparabacrim_cede <- as.numeric(seq(min(df$ln -
    atparabacrim_cede, na.rm = TRUE),
22
    max(df$ln_atparabacrim -
23
    length.out = 100))
24
25 # predict new data using model_7 on the prediction data created
26 preds_para <- predict(model_7, newdata = prediction_data_para, type = "
    response",
27
    interval = "confidence")
28
29 # unlist predicted column
30 prediction_data_para$predicted <- preds_para[, "fit"]
31
32 # confidence intervals around predicted probabilities
33 prediction_data_para$lower_95 <- preds_para[, "lwr"]
34 prediction_data_para$upper_95 <- preds_para[, "upr"]
35
36 # plot predicted % Yes votes in the referendum against farc attacks
37 pdf("pred_plot_para.pdf")
38 pred_plot_para <- ggplot(prediction_data_para, aes(x = exp(ln_atparabacrim -
    cede), y = predicted)) +
39   geom_ribbon(aes(ymin=lower_95, ymax=upper_95), alpha = 0.2) +
40   geom_point() +
41   geom_rug(data = df, aes(x=exp(ln_atparabacrim_cede)),
42     inherit.aes = FALSE, alpha = 0.4, sides = "b") +
43   labs(y = "Predicted % Yes Votes in Referendum",
44     x = "Para/BACRIM",
45     title = "Para/BACRIM Attacks vs Predicted Votes in Referendum") +
46   theme_minimal()
47 pred_plot_para

```

Figures 2 and 3 clearly illustrate that the interpretation of the OLS coefficients made by the authors is incorrect. The lines illustrate a logarithmic curve in the relationship between the FARC attack counts and the % yes votes in the referendum, as well as the para/BACRIM attack counts and the % yes votes in the referendum. This indicates that the *Beta* coefficient

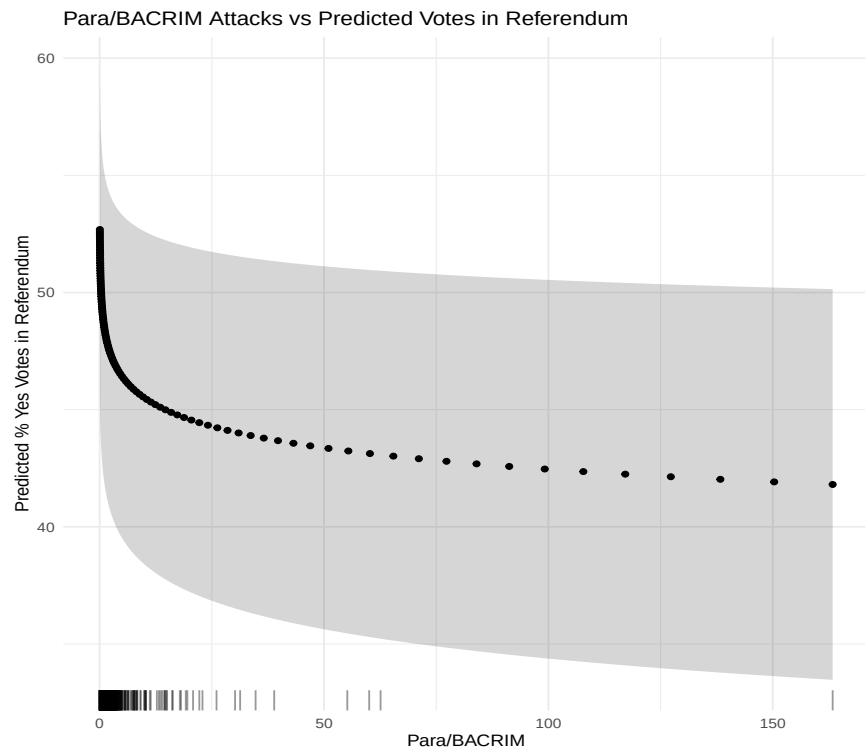


Figure 3: Para/BACRIM Attack Counts vs Predicted Probabilities of % Yes Votes in Referendum

for the impact of the log transformation of these attacks needs to be divided by 100. The true interpretation for the (ln) **FARC attacks** and (ln) **Para/BACRIM** coefficients are:

FARC Attacks: Holding all other variables constant, for every 1% increase in the mean log of FARC attacks in a given municipality, the yes vote in the referendum increases by approximately 0.0106 points.

Para/BACRIM Attacks: Holding all other variables constant, for every 1% increase in the mean log of para/BACRIM attacks in a given municipality, the yes vote in the referendum decreases by approximately 0.0132 points.

Within these figures, I also plotted the 95% confidence intervals, which are incredibly large, but do not vary among varying numbers of FARC or para/BACRIM attacks due to setting all other covariates at their mean to generate the predicted % yes vote in the referendum. These confidence intervals do illustrate the fact that municipalities with very high and very low numbers of FARC or para/BACRIM attacks could ultimately have the same % yes votes in the referendum, due to the fact that the confidence intervals at both high and low counts of FARC and para/BACRIM attacks include the same values. I chose to add a rug plot along the x-axis to illustrate the distribution of the true number of FARC and para/BACRIM attacks, which can be seen as incredibly skewed to the right, indicating that in most municipalities, and subsequently departments, there were very few numbers of FARC or para/BACRIM attacks.

With all of this being considered, the authors might have chosen to log the counts of FARC and para/BACRIM attacks to create a more uniform distribution and subsequently make their results interpretable using an OLS regression. It is regrettable that this subsequently makes their interpretation incorrect, and resultingly significantly decreases the magnitude of their findings.

References

- Gelman, A., & Hill, J. (2018). *Data analysis using regression and multilevel/hierarchical models*. Cambridge University Press.
- Clark, T. S., & Linzer, D. A. (2014). Should I use fixed or random effects? *Political Science Research and Methods*, 3(2), 399–408. <https://doi.org/10.1017/psrm.2014.32>
- Clark, M. (n.d.). *Models demystified – Michael Clark*. Github.io. Retrieved March 25, 2026, from <https://m-clark.github.io/#category=mixed%20models>